**CHAPTER 11**

**INTERMOLECULAR FORCES**

1. The boiling point of $CH_4$ is much lower than that of HF. This is because:
   A. of ion-dipole interactions in $CH_4$.
   B. of dipole-dipole interactions in $CH_4$.
   C. of hydrogen bonding in HF.
   D. $CH_4$ is polar.

2. For a given substance, which of the following phase transitions should *RELEASE* the most energy.
   A. Liquid to gas
   B. Gas to solid
   C. Gas to liquid
   D. Solid to gas

3. At room temperature, $F_2$ and $Cl_2$ are gases, $Br_2$ is a liquid, and $I_2$ is a solid. This is because:
   A. dipole-induced dipole interactions increase with molecular size.
   B. polarity increases with molecular size.
   C. dipole-dipole interactions increase with molecular size.
   D. dispersion interactions increase with molecular size.

4. What type of intermolecular forces holds liquid $N_2$ together?
   A. ionic bonding
   B. London forces
   C. hydrogen bonding
   D. dipole-dipole interaction

5. Which response includes only those compounds that can exhibit hydrogen bonding?
   A. $AsH_3$, $H_2Te$
   B. $AsH_3$, $CH_3NH_2$
   C. $CH_4$, $AsH_3$, $H_2Te$
   D. $CH_3NH_2$, HF

6. Which of the following boils at the highest temperature?
   A. $CH_4$
   B. $C_2H_6$
   C. $C_3H_8$
   D. $C_4H_{10}$

7. Which probably has the lowest boiling point at 1.00 atm?
   A. HF
   B. HCl
   C. HBr
   D. HI

8. The normal boiling point of a liquid is
   A. the temperature at which the vapor pressure equals 760 torr.
   B. the temperature above which the substance cannot exist as a liquid regardless of the pressure.
   C. the only temperature at which there can be equilibrium between liquid and gas.
   D. the temperature at which the liquid will usually boil.

9. Which of the following phase changes is(are) endothermic?

   1. melting
   2. vaporization
   3. sublimation
   4. condensation
   5. deposition
   6. freezing

   A. 1, 2, and 3
   B. 4, 5, and 6
   C. 1 and 2 only
   D. 4 and 6 only

10. Which one of the following elements is least likely to participate in a hydrogen bond?
    A. O
    B. F
    C. S
    D. N

11. Which of the following compounds has the highest boiling point?
    A. $CH_4$
    B. $CH_3CH_3$
    C. $CH_3CH_2OH$
    D. $CH_3OCH_3$

12. Which of the following best describes all the intermolecular forces exhibited by a pure sample of $PH_3$?
    A. dispersion, dipole-dipole, and hydrogen bonding
    B. dispersion only
    C. dispersion and hydrogen bonding
    D. dispersion and dipole-dipole

13. Which of the following best describes all the intermolecular forces exhibited by a pure sample of $CH_3NH_2$?
    A. dispersion, dipole-dipole, and hydrogen bonding
    B. dispersion only
    C. dispersion and hydrogen bonding
    D. dispersion and dipole-dipole

14. Which of the following best describes all the intermolecular forces exhibited by a pure sample of $CS_2$?
    A. dispersion, dipole-dipole, and hydrogen bonding
    B. dispersion only

C. dispersion and hydrogen bonding
D. dispersion and dipole-dipole

15. Which of the following best describes all the intermolecular forces exhibited by a pure sample of $CH_3Cl$?
A. dispersion, dipole-dipole, and hydrogen bonding
B. dispersion only
C. dispersion and hydrogen bonding
D. dispersion and dipole-dipole

16. Which molecule would be expected to have the highest viscosity?
A. $HOCH_2CH_2OH$
B. $CH_3OCH_3$
C. $H_2O$
D. $C_2H_5OH$

17. Which pure substance would have the smallest molar heat of vaporization?
A. $C_4H_{10}$
B. $C_5H_{12}$
C. $C_3H_8$
D. $CH_4$

18. Which molecule would be expected to have the highest vapor pressure?
A. $H_2O$
B. $CH_3OCH_3$
C. $CH_3CH_2CH_2CH_2OH$
D. $C_2H_5OH$

19. The correct order of increasing boiling point is
A. He < $C_2H_5OH$ < $C_2H_6$ < KBr
B. KBr < $C_2H_6$ < $C_2H_5OH$ < He
C. He < $C_2H_6$ < $C_2H_5OH$ < KBr
D. $C_2H_5OH$ < $C_2H_6$ < He < KBr

20. For a pure substance, the $\Delta H_{freezing}$ and $\Delta H_{deposition}$ are known to be -15 kJ/mol and -75 kJ/mol, respectively. What is $\Delta H_{vaporization}$ for the substance?
A. 60 kJ/mol
B. - 60 kJ/mol
C. 90 kJ/mol
A. -90 kJ/mol

21. The correct order of the **viscosity is ---------.** All are at the same 50ºC temperature.
I. $CH_3CH_2OCH_3$
II. $CH_3(CH_2)_2NH_2$
III. $CH_3(CH_2)_4NH_2$
IV. $CH_3(CH_2)_2CH_3$
A. I > II > III > IV
B. III > II > I > IV
C. IV > II > I > III
D. IV > I > II > III

22. The correct order of the **surface tension is ---------.** All are at the same 50ºC temperature.
I. $CH_3CH_2OCH_3$
II. $CH_3(CH_2)_2NH_2$
III. $CH_3(CH_2)_4NH_2$
III. $CH_3(CH_2)_2CH_3$
A. I > II > III > IV
B. III > II > I > IV
C. IV > II > I > III
D. IV > I > II > III

23. Which of the following liquids would have the lowest viscosity, factoring in both the impact of the substance and the temperature?
A. $C_3H_7OH$ at 25°C
B. $C_3H_7OH$ at 75°C
C. $MgBr_2$ at 25°C
D. $C_5H_{11}OH$ at 75°C

24. Predict which of the following liquid/temperature scenarios would have the **HIGHEST** vapor pressure and the **LOWEST** surface tension?
A. $C_6H_{14}$ at 275 K
B. $C_6H_{14}$ at 299 K
C. $C_5H_{12}$ at 299 K
D. $HOC_4H_8OH$ at 299 K

25. At room temperature, the vapor pressure pattern is acetone > heptane > ethanol. Which **one** of the following statements is **FALSE**:
A. a substance with higher vapor pressure is held together by weaker binding forces
B. ethanol has the lowest vapor pressure and strongest intermolecular force due to hydrogenbonding
C. heptane has lower vapor pressure than acetone due to London dispersion forces
D. acetone would have a higher boiling point than heptane

**Answer Key**

| Question # | Answer | Question # | Answer | Question # | Answer | Question # | Answer |
|---|---|---|---|---|---|---|---|
| 1 | C | 7 | B | 13 | A | 19 | C |
| 2 | B | 8 | A | 14 | B | 20 | A |
| 3 | D | 9 | A | 15 | D | 21 | B |
| 4 | B | 10 | C | 16 | A | 22 | B |
| 5 | D | 11 | C | 17 | D | 23 | B |
| 6 | D | 12 | D | 18 | A | 24 | C |